

2-10 Fluently Add within 1,000

Learning goal: I can add 3-digit numbers using partial sums or by adjusting addends, and choose a strategy that works well for me.

Textbook: pages 71–74. The book adds $722 + 169$ with partial sums (row and column) and by adjusting to $721 + 170$.

Words to Know

addend / sum	A number you add / the answer when you add.
partial sums	Add the hundreds, tens, and ones separately, then add those answers together.
adjust addends	Take some from one addend and give it to the other so a number is friendlier.
strategy	A way to solve a problem.
efficient	Quick and easy, and still correct.
extra information	A number in a story that you do not need to answer the question.

How to Teach It

- 1. Partial sums:** add hundreds + hundreds, tens + tens, ones + ones. Then add the three partial sums. Write them in a row or in a column.
- 2. Adjust addends:** if a number is close to a ten or hundred (like 259 or 299), take a little from the other addend and give it to that number.
- 3.** If you change only one addend, undo it at the end. For $336 + 457$, change 336 to 340: $340 + 457 = 797$. That is 4 too many, so the sum is 793.
- 4.** Let your child choose. Any correct strategy is fine. Ask which one felt easier and why.
- 5.** In stories, read the question first and cross out numbers you do not need.

Tip: Ask, “Is one number close to a ten or a hundred?” If yes, adjusting is quick. Partial sums always work.

Worked Examples

Example 1: Partial sums in a row

$$\begin{array}{l}
 634 + 259 = 893 \\
 600 + 200 = 800 \\
 30 + 50 = 80 \\
 4 + 9 = 13 \\
 800 + 80 + 13 = \mathbf{893}
 \end{array}$$

Example 2: Partial sums in a column

$$\begin{array}{r}
 634 \\
 + 259 \\
 \hline
 600 + 200 = 800 \\
 30 + 50 = 80 \\
 4 + 9 = 13 \\
 \hline
 \mathbf{893}
 \end{array}$$

Example 3: Adjust addends

$$\begin{array}{rcl}
 634 & + & 259 = ? \\
 \mathbf{-1} & & \mathbf{+1} \\
 \downarrow & & \downarrow \\
 633 & + & 260 = \mathbf{893}
 \end{array}$$

Take 1 from 634 and give it to 259. The sum stays the same.

Example 4: Find the useful information

A school of 410 students collected 356 books in May and 218 in June. How many books in all?

The 410 is extra. $356 + 218 \rightarrow 354 + 220 = \mathbf{574}$ books.

Skills Checklist *Check off each skill your child can do without help.*

- | | |
|--|---|
| ☐ Add with partial sums in a row. | ☐ Fix the sum after changing only one addend. |
| ☐ Add with partial sums in a column. | ☐ Choose and explain a strategy. |
| ☐ Adjust addends and keep the sum the same. | ☐ Ignore extra information and solve two-step stories. |

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Name _____ Date _____

Together *Work with your grown-up. Talk about each step.*

- A.** Find $538 + 246$ two ways.

Partial sums

Adjust addends

Sum: _____

- B.** Rosa and Ben find $457 + 238$. Rosa adjusts the addends. Ben uses partial sums. Can either strategy be used? Explain. What is the sum?

Sum: _____

- C.** A farm grew 274 pumpkins and 419 squash. It sold 300 vegetables at the market. How many vegetables did the farm grow? Which number do you not need?

Vegetables grown: _____

Number not needed: _____

On My Own *Try these by yourself. Show your work.*

- 1.** Solve. Use any strategy.

$$546 + 239 = \underline{\hspace{2cm}}$$

- 2.** Solve. Use any strategy.

$$\underline{\hspace{2cm}} = 152 + 637$$

3. Leo adds $436 + 285$. He adds 4 to 436 and finds $440 + 285 = 725$. Then he subtracts 4 from the sum. Do you agree with his strategy? Explain.

Sum: _____

4. Ana adds $263 + 518$ by writing the addends in a row. Show another way to find $263 + 518$.

Sum: _____

5. A bike path is 342 meters long. Kim rides the whole path 2 times. How many meters does she ride?

Meters: _____

6. A reading club read 215 books in fall and 162 books in winter. In spring, they read 118 more books than in fall and winter together. How many books did they read in spring?

Books: _____

7. Solve. Explain your strategy.
 $368 + 257 =$ _____

Strategy: _____

8. Solve. Explain your strategy.
 $526 + 297 =$ _____

Strategy: _____

9. Ms. Lee needs 700 stickers. Stickers come in packs of 250. How many packs should she buy? Explain.

Packs: _____

10. A class collected 356 cans in March and 389 cans in April. They gave 500 cans to a food bank. How many cans did the class collect?

Cans: _____

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Together

A	Partial sums: $500 + 200 = 700$; $30 + 40 = 70$; $8 + 6 = 14$; $700 + 70 + 14 = \mathbf{784}$ Adjust: $538 + 246 \rightarrow 540 + 244 = \mathbf{784}$ <i>Any correct adjustment is fine (e.g., $534 + 250 = 784$).</i>
B	Yes, both work. Adjusting: $455 + 240 = 695$. Partial sums: $600 + 80 + 15 = 695$. Sum: 695 <i>Partial sums adds every place. Adjusting moves 2 from one addend to the other, so the total does not change.</i>
C	$274 + 419 = \mathbf{693}$ vegetables. The 300 is not needed.

On My Own

1	785 <i>Items 1–2: accept any correct strategy (e.g., $545 + 240 = 785$).</i>
2	789
3	Yes. Adding 4 made the sum 4 too big, so subtracting 4 fixes it: $725 - 4 = \mathbf{721}$.
4	Sample (stacked partial sums): $200 + 500 = 700$; $60 + 10 = 70$; $3 + 8 = 11$; sum 781 . <i>Also accept adjusting: $261 + 520 = 781$, or any correct other way.</i>
5	$342 + 342 = \mathbf{684}$ meters
6	$215 + 162 = 377$. $377 + 118 = \mathbf{495}$ books.
7	625 <i>Sample: $600 + 110 + 15 = 625$, or $370 + 255 = 625$. Any explained strategy.</i>
8	823 <i>Sample: adjust to $523 + 300 = 823$.</i>
9	$250 + 250 = 500$ is not enough. $500 + 250 = 750$ is enough. 3 packs . <i>2 packs is a common wrong answer ($500 < 700$).</i>
10	$356 + 389 = \mathbf{745}$ cans. The 500 is extra information.